

Question	Marking Guidance	Mark	Comments
03.1	For 1 mark, accept any two from Prokaryotes have No membrane-bound organelles/correct example OR (Single,) circular/loop DNA (in cytoplasm) OR DNA free in cytoplasm OR DNA not associated with proteins/histones OR Murein/peptidoglycan (in) cell wall;	1 (AO1)	Apply list rule Accept (prokaryotes) only have smaller ribosomes /60S/70S Accept mesosome Accept no introns Accept nucleoid for single, circular DNA Reject nucleosome Reject plasmid Reject (bacterial) chromosome Reject capsule/slime layer Reject flagellum

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03.2	Hydrogen ☒	1 (AO1)	

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03.3	1. Hydrophobic side next to/in/face fatty acids/tails OR Hydrophobic side next to/in/face hydrophobic (part of) phospholipid/bilayer; 2. Hydrophilic sides allow ion movement through membrane OR Hydrophilic sides form a channel;	2 (2 x AO2)	1 and 2 Accept 'part/region/bit/half' for side 2. Accept water OR charged/polar molecules/substances OR water-soluble substances for ions

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03.4	Correct answer for 2 marks, 2.5;; Accept for 1 mark, 2.5434 (Correct answer but 2 or more decimal places) OR 10.2 (correct area calculation using diameter, to 1 decimal place) OR 6.6 (correct calculation using radius $(4 - 1.1) \div 2$, to 1 decimal place) OR 26.4 (correct calculation using diameter, $4 - 1.1$, to 1 decimal place)	2 (2 x AO2)	

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03.5	1. Cholesterol stabilises (the membrane) OR Cholesterol restricts the movement of molecules/phospholipids/fatty acid (tails) (making up the membrane); 2. (So) APs do not make channels in (eukaryotic) membranes OR (So) APs cannot enter the (eukaryotic) membrane;	2 (2 x AO3)	1. Accept makes (membrane) less flexible OR less fluid OR stiffer OR rigid OR gives structural support for stabilises 1. Ignore strength 1. Accept holds together for restricts movement 2. Accept fewer for do not 2. Accept cannot fit OR be positioned OR sit in OR form in OR embed OR disrupt for enter
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03.6	1. Antibody binds to AP OR Gold (present) where AP located; 2. (As antibody/tertiary structure is) <u>complementary</u> (to AP); 3. Gold interacts with electrons (in TEM); 4. (T)EM (used as it) has a high resolution;	3 max (3 x AO2)	1. Accept attaches OR forms antibody-AP OR antibody-antigen complex, for binds 2. Reject reference to active site 3. For 'interact' accept scattered/deflected/reflected OR blocked/absorbed/ bounced/ interrupted OR a description of these

